



Rujul Matta
Bachelor of Technology
in Engineering Physics
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EDUCATION

Degree	Institute/Board	CGPA/Percentage	Year
Bachelor of Technology	Indian Institute of Technology, Ropar	7.47 (Till 4th Sem)	2024-2028
Senior Secondary	Central Board of Secondary Education	92.8%	2024
Secondary	Central Board of Secondary Education	95.4%	2022

PROJECTS

- **Meridian: Decentralized Lending Protocol** Ongoing
Self Directed – Solana / DeFi GitHub
 - Built a **Solana-based decentralized lending protocol** in **Rust** with **Anchor**, **SPL Token-2022** for scalable and secure on-chain lending.
 - Implemented a **permissionless liquidation and yield distribution mechanism** with automated collateral monitoring, oracle checks, reserve buffering, and protocol solvency protections.
 - Engineered secure smart contract workflows using **PDA**s, **CPI**s, fixed-point math, event-driven logic, and **multisig controls** to ensure reliability and fault tolerance.
- **Multithreaded HTTP Web Server** 2026
Self Directed – Rust Systems Programming GitHub
 - Built a **multithreaded HTTP web server** in **Rust** using **TCP sockets**, capable of accepting concurrent client connections, parsing HTTP requests, and serving static content with valid HTTP responses.
 - Implemented a reusable **thread pool** to replace the thread-per-request model, improving scalability, reducing thread creation overhead, and enabling efficient handling of multiple simultaneous client requests.
 - Applied core **systems programming** concepts including **network programming**, **socket communication**, **HTTP protocol**, **ownership**, and **concurrency**, strengthening low-level software engineering and performance optimization skills.
- **Link Safety Extension** 2026
Cybersecurity GitHub
 - Developing a **browser extension** in TypeScript that analyses URLs in real time, cross-referencing against threat databases and heuristic pattern detection to flag malicious or phishing links before the user visits them.
 - Integrated a **Python-based classification pipeline** using Scikit-learn to train a model on known malicious URL datasets, enabling the extension to detect suspicious patterns beyond static blacklists.
 - Designed with a focus on **lightweight performance and user privacy**, ensuring zero data is sent to external servers – all analysis runs client-side or via a local backend.
- **Centralized Cryptocurrency Exchange** 2026
Self Directed – Rust Backend Development GitHub
 - Developed a **centralized cryptocurrency exchange** in **Rust**, implementing secure user authentication, wallet management, and RESTful APIs for trading operations.
 - Built a high-performance **order matching engine** using an in-memory order book with **price-time priority**, supporting market and limit orders for efficient trade execution.
 - Applied **concurrency**, transaction validation, persistent storage, and role-based access control to build a secure and scalable backend for financial applications.

TECHNICAL SKILLS

- **Programming Languages:** Rust, C++, Python, TypeScript, JavaScript
- **Systems & Networking:** RESTful API Design, Multithreading, Concurrent Programming, TCP/IP, HTTP, Socket Programming, Linux
- **Databases & DevOps:** SQL, PostgreSQL, MongoDB, Docker
- **Blockchain:** Solana, Anchor, Smart Contracts, SPL Token-2022, PDAs, CPIs, LiteSVM, Unit Testing
- **Data Structures & Algorithms:** Arrays, Trees, Graphs, Dynamic Programming, Hashing, Recursion
- **Libraries & Frameworks:** Scikit-learn, NumPy, Pandas
- **Tools:** Git, GitHub, Docker, Arduino, ESP32

KEY COURSES TAKEN

- **Physics & Mathematics:** Applied Mathematical Physics, Calculus, Linear Algebra, Probability and Stochastic Processes
- **Engineering & Computing:** Signals and Systems, Digital Circuits, Basic Electronics, Circuit Theory, Control Engineering, Introduction to Computing and Data Structures

POSITIONS OF RESPONSIBILITY

- **Secretary**, Softcom - Software Community, IIT Ropar 2026 – Present
- **Representative**, Softcom - Software Community, IIT Ropar 2025 – 2026
- **Executive**, FinCom, IIT Ropar 2024 – 2025